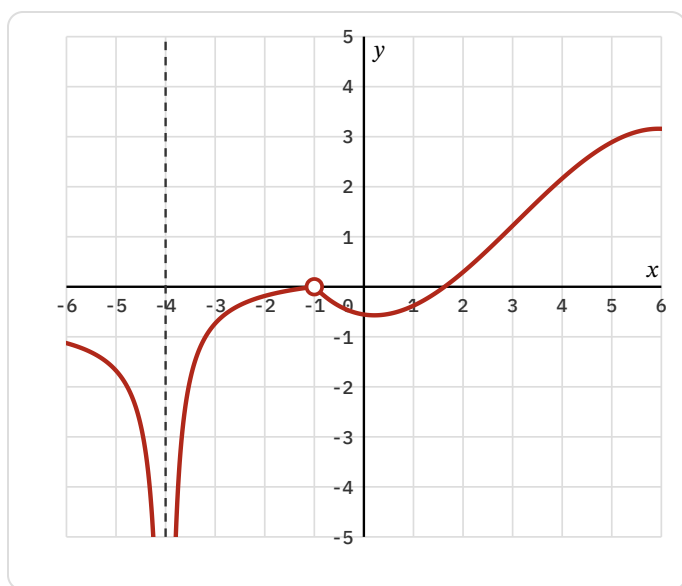


Mock Midterm A

Practice midterm (AI-Free) · Sections 1.1–1.4

Directions: Answer each question below. Be sure to show work to receive full credit. Work this exam on your own, without notes or AI, the way you will on the proctored midterm. Be sure the notation in your solutions matches the notation from the lesson videos. Write **DNE** if a value or limit does not exist. Suggested time: 75 minutes.

1. Use the graph of $y = f(x)$ to state the value of each quantity, if it exists. If it does not exist, write DNE. 8 pts



- a.) $f(-4)$ _____ (1)
- b.) $\lim_{x \rightarrow -4^-} f(x)$ _____ (1)
- c.) $\lim_{x \rightarrow -4^+} f(x)$ _____ (1)
- d.) $\lim_{x \rightarrow -4} f(x)$ _____ (1)
- e.) $f(-1)$ _____ (1)
- f.) $\lim_{x \rightarrow -1^-} f(x)$ _____ (1)
- g.) $\lim_{x \rightarrow -1^+} f(x)$ _____ (1)
- h.) $\lim_{x \rightarrow -1} f(x)$ _____ (1)

2. Let $f(x) = \frac{\sqrt{x}-4}{x-16}$. We want to know what happens to the outputs when x is near **16**.

7 pts

x	15.9	15.99	15.999	16	16.001	16.01	16.1
$f(x)$	0.125196	0.12502	0.125002	?	0.124998	0.12498	0.124805

a.) Can you find $\lim_{x \rightarrow 16} f(x)$ by plugging in $x = 16$? (1)

(A) No — substituting gives $\frac{0}{0}$, which is undefined

(B) Yes — just compute $f(16)$

(C) No — f is undefined for every x near **16**

b.) As $x \rightarrow 16^-$, $f(x)$ approaches... (2)

c.) As $x \rightarrow 16^+$, $f(x)$ approaches... (2)

d.) $\lim_{x \rightarrow 16} f(x) =$ (2)

3. Suppose

4 pts

$$\lim_{x \rightarrow 4} f(x) = 1, \quad \lim_{x \rightarrow 4} g(x) = -4, \quad \lim_{x \rightarrow 4} h(x) = 25$$

Use the properties of limits to find each limit.

a.) $\lim_{x \rightarrow 4} ([f(x)]^3 - g(x))$ (1)

b.) $\lim_{x \rightarrow 4} \sqrt{h(x)} f(x)$ (1)

c.) $\lim_{x \rightarrow 4} \frac{g(x)-2}{f(x)}$ (1)

d.) $\lim_{x \rightarrow 4} x^2 f(x)$ (1)

4. Calculate each limit algebraically. (Direct substitution gives $\frac{0}{0}$ — simplify first.) If it doesn't exist, write DNE. 4 pts

a.) $\lim_{x \rightarrow 25} \frac{\sqrt{x}-5}{x-25}$ _____ (2)

b.) $\lim_{h \rightarrow 0} \frac{\frac{1}{4+h} - \frac{1}{4}}{h}$ _____ (2)

5. Compute each limit algebraically. Use ∞ / $-\infty$ for unbounded results. 6 pts

a.) $\lim_{x \rightarrow \infty} \frac{-3x^3 - x^2}{5x}$ _____ (2)

b.) $\lim_{x \rightarrow \infty} \frac{7x+2}{-5x-5}$ _____ (2)

c.) $\lim_{x \rightarrow \infty} \frac{3x^3}{-5x^4 - 5x^3 + 7x^2}$ _____ (2)

6. Let

7 pts

$$G(x) = \begin{cases} x^2 - 2 & x < -2 \\ -x & x > -2 \end{cases}$$

Find each quantity. If it does not exist, write DNE.

a.) $G(-2)$ _____ (1)

b.) $\lim_{x \rightarrow -2^-} G(x)$ _____ (1)

c.) $\lim_{x \rightarrow -2^+} G(x)$ _____ (1)

d.) $\lim_{x \rightarrow -2} G(x)$ _____ (1)

e.) $G(1)$ _____ (1)

f.) $\lim_{x \rightarrow -\infty} G(x)$ _____ (1)

g.) $\lim_{x \rightarrow \infty} G(x)$ _____ (1)

7. Let

6 pts

$$f(x) = \begin{cases} \frac{2x^2 - 5x + 3}{2x - 3} & x < \frac{3}{2} \\ 4x + 4 & x \geq \frac{3}{2} \end{cases}$$

Find each limit or function value algebraically. Write DNE if it doesn't exist.

a.) $\lim_{x \rightarrow 5} f(x)$ _____ (1)

b.) $\lim_{x \rightarrow -2} f(x)$ _____ (1)

c.) $f\left(\frac{3}{2}\right)$ _____ (1)

d.) $\lim_{x \rightarrow \frac{3}{2}^-} f(x)$ _____ (1)

e.) $\lim_{x \rightarrow \frac{3}{2}^+} f(x)$ _____ (1)

f.) $\lim_{x \rightarrow \frac{3}{2}} f(x)$ _____ (1)

8. Is $f(x)$ continuous at $x = -1$?

5 pts

$$f(x) = \begin{cases} \frac{x^2 + 6x + 5}{x + 1} & x \neq -1 \\ 4 & x = -1 \end{cases}$$

a.) $f(-1)$ _____ (2)

b.) $\lim_{x \rightarrow -1} f(x)$ _____ (2)

c.) Is f continuous at $x = -1$? (1)

(A) Yes — $f(-1)$ exists, the limit exists, and they are equal

(B) No — the one-sided limits differ, so the limit does not exist

(C) No — the limit exists but does not equal $f(-1)$

(D) No — $f(-1)$ is undefined

9. Classify each function at the given x -value.

3 pts

a.) $f(x) = x^2 + 5x - 4$ at $x = -1$ (1)

(A) Continuous

(B) Removable discontinuity

(C) Jump discontinuity

(D) Infinite discontinuity

b.) $f(x) = \frac{x^2-4}{x+2}$ at $x = -2$ (1)

(A) Continuous

(B) Removable discontinuity

(C) Jump discontinuity

(D) Infinite discontinuity

c.) $f(x) = \frac{x+2}{x^2+x-2}$ at $x = -2$ (1)

(A) Continuous

(B) Removable discontinuity

(C) Jump discontinuity

(D) Infinite discontinuity

10. Let $C(q) = q^2 + 3q + 200$, where $C(q)$ represents the cost, in dollars, of producing q items. 7 pts
Find the average rate of change over $[4, 8]$.

a.) $C(4)$ _____ (2)

b.) $C(8)$ _____ (2)

c.) Average rate of change (number only) _____ (2)

- d.) Which sentence correctly states the result **with units**? (1)

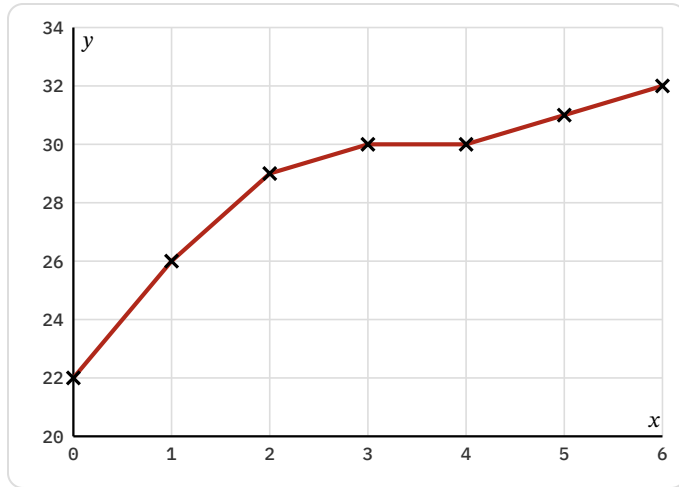
(A) At exactly $q = 8$ items, the cost is changing at a rate of **15** dollars per item.

(B) Between $q = 4$ and $q = 8$ items, the cost changes at an average rate of **15** dollars per item.

(C) Between $q = 4$ and $q = 8$, the cost changes at an average rate of **15** items per dollar.

(D) The cost changes by a total of **15** dollars from $q = 4$ to $q = 8$.

11. The graph shows the closing price (in dollars) of a stock at the start of each month in 2026, where $x = 0$ is January, $x = 1$ is February, and so on. 3 pts



- a.) Average rate of change on $[2, 5]$ (dollars per month) _____ (2)
- b.) On which interval is the average rate of change greatest? (1)
- (A) $[0, 2]$
- (B) $[2, 4]$
- (C) $[4, 6]$

12. Let $f(x) = 2x^2 + 2x + 3$. Find and simplify the difference quotient $\frac{f(x+h)-f(x)}{h}$, then use it to fill in the table. 7 pts

a.) Simplified difference quotient (in terms of x and h) _____ (3)

b.) $x = 3, h = 1$ _____ (1)

c.) $x = 3, h = 0.1$ _____ (1)

d.) $x = 3, h = 0.01$ _____ (1)

e.) As $h \rightarrow 0$, what value do these approach? _____ (1)

13. Let $f(x) = x^3 - x$. Use the limit definition $f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$ to find $f'(x)$. 8 pts

a.) $f'(x) =$ _____ (3)

b.) $f'(-2) =$ _____ (2)

c.) Slope of the tangent line to f at $x = 2$ _____ (2)

d.) What does $f'(-2) = 11$ mean? (1)

(A) The instantaneous rate of change of f at $x = -2$ is 11 — the slope of the tangent line there.

(B) The value of f at $x = -2$ is 11.

(C) The slope of the secant line from $x = -2$ to $x = -2 + 1$ is 11.

(D) The average rate of change of f from $x = 0$ to $x = -2$ is 11.

14. Let $f(x) = -x^2 + x$. Find the equation of the tangent line to the graph of f at $x = -2$. 7 pts

a.) $f(-2)$ _____ (2)

b.) $f'(-2)$ _____ (2)

c.) Tangent line _____ (3)

15. Let $T(t) = -t^2 + 30t + 70$ be the temperature, in °F, of an oven t minutes after it is turned on. 6 pts

a.) $T'(t) =$ _____ (3)

b.) $T'(12)$ _____ (2)

c.) Interpret $T'(12) = 6$. (1)

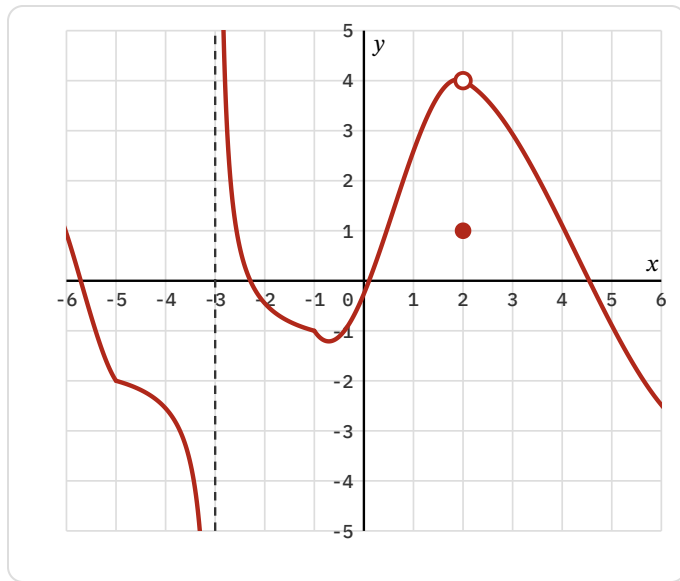
(A) At $t = 12$ minutes, the oven's temperature is increasing at a rate of 6 degrees Fahrenheit per minute.

(B) At $t = 12$ minutes, the oven's temperature is decreasing at a rate of 6 degrees Fahrenheit per minute.

(C) At $t = 12$ minutes, the oven's temperature is 6 degrees Fahrenheit.

(D) Over the first 12 minutes, the oven's temperature changed at an average rate of 6 degrees Fahrenheit per minute.

16. At each labeled x -value, decide whether the graph of f is continuous there and whether it is differentiable there. 4 pts



- a.) At $x = -5$ (1)
(A) Continuous and differentiable
(B) Continuous but NOT differentiable
(C) Not continuous (so not differentiable)
- b.) At $x = -3$ (1)
(A) Continuous and differentiable
(B) Continuous but NOT differentiable
(C) Not continuous (so not differentiable)
- c.) At $x = -1$ (1)
(A) Continuous and differentiable
(B) Continuous but NOT differentiable
(C) Not continuous (so not differentiable)
- d.) At $x = 2$ (1)
(A) Continuous and differentiable
(B) Continuous but NOT differentiable
(C) Not continuous (so not differentiable)